

PAVAN KUMAR ETTA

BOSTON, MA | +1 (508)-863-0500 | pavankumar.etta02@gmail.com | [GITHUB](#) | [LINKEDIN](#) | [GOOGLE SCHOLAR](#)

PROFESSIONAL SUMMARY

Software Engineer with 4+ years building scalable distributed systems, cloud-native microservices, and real-time data pipelines across AWS/GCP. Reduced latency **40%**, query time **72%**, and ingestion throughput **35%** while shipping reliable backend platforms for high-volume, production-grade engineering teams. Strong in software engineering, APIs, event-driven architecture, Kubernetes, databases, and performance optimization for high-scale software systems.

WORK EXPERIENCE

- Graduate Research Assistant | University of Massachusetts, Dartmouth | USA** **Nov 2024 – Dec 2025**
- Conducting research in medical data systems at UMass Dartmouth, focusing on high-performance database design and clinical data visualization for real-time health analytics applications.
 - Redesigned MySQL schema architecture using partitioning and compound indexing, reducing query response times by **70%** under high concurrent load across medical research datasets.
 - Built interactive visualization dashboards using React, TypeScript, and Plotly.js, enabling real-time exploration of clinical time-series data, trend analysis, anomaly detection, and faster decision-making for healthcare teams.
 - Developed and evaluated LSTM-based predictive models for time-series health data (see Water Quality Analysis project), achieving MAE of 1.45 on **30GB+** environmental datasets processed via PySpark on GCP.
 - Containerized full application stack with Docker, improving environment reproducibility and increasing platform adoption by **25%**.

- Software Engineer II | Rhenix Lifesciences LLP | India** **Aug 2021 – Aug 2024**
- Shipped a real-time health monitoring platform for **2,000+** wearable device users, enabling clinicians to track patient vitals with sub-second data freshness and powering live patient alerts processing **50K+** daily health events with guaranteed delivery.
 - Built the full backend API layer for iOS and Android health monitoring apps used by **2,000+** active users, owning end-to-end feature delivery from schema design to production deployment while maintaining **99.9%** uptime SLAs.
 - Reduced end-to-end system latency by **40%** by implementing circuit breakers, parallel workflows, and optimized scheduling, improving reliability, fault tolerance, and real-time app responsiveness for end users.
 - Developed a centralized analytics dashboard processing **50K+** daily health metrics, giving clinical teams actionable visibility into patient data and reducing data preprocessing time by **40%** through optimized NumPy/Pandas pipelines.
 - Led integration of event-driven async workflows using AWS MSK (Kafka), enabling reliable multi-service communication across the product backend and supporting horizontal scaling as the user base grew.

- Software Engineer | Cure Science Institute | USA** **Jun 2021 – Aug 2021**
- Built real-time ingestion pipelines processing embedded hardware telemetry, improving signal quality and system stability by **30%**.
 - Implemented an ML-based anomaly detection system using Scikit-learn (**93%** accuracy), automating equipment health monitoring.

- Software Engineer | Maxgen Technologies | India** **Jun 2019 – June 2021**
- Engineered backend services for a high-traffic B2B platform handling **100K+** daily transactions, owning core feature development across a distributed architecture with **99.99%** uptime.
 - Refactored a legacy monolithic product into containerized microservices (Docker/Kubernetes), cutting deployment cycles by **50%**.
 - Optimized SQL query execution and implemented multi-level Redis caching for frequently accessed product data, reducing database read latency by **30%** and improving response times for end users.
 - Built automated testing suites (JUnit/Mockito) achieving **85%** code coverage, reducing production bug reports by **25%**.

SKILLS

Programming: Python, JavaScript, Typescript, C/C++, Java, Go

Backend & APIs: FastAPI, RESTful APIs, gRPC, GraphQL, Event-Driven Architecture, System Design, Postman API

Data Engineering: Apache Spark, PySpark, Airflow, Snowflake, Hadoop, dbt

Databases & Caching: PostgreSQL, MySQL, DynamoDB, MongoDB, Cassandra, Redis, Elasticsearch

Cloud & Infra: AWS (Lambda, EC2, S3, RDS, MSK, EKS), GCP, Azure, Terraform, Serverless

DevOps & Observability: Docker, Kubernetes, Helm, CI/CD, GitHub Actions, GitLab CI, Jenkins, Prometheus, Grafana, Datadog

PROJECTS

- House Price Prediction** [\[GITHUB\]](#) (AWS, Django, Python)
- Utilized AWS Sagemaker, EC2, EBS, Lambda, Lex, and API Gateway to develop and deploy a Django app. Built a Lex chatbot for data collection and a Sagemaker model for price prediction with **90%** accuracy.

- Water Quality Analysis** [\[GITHUB\]](#) (NetCDF, Tableau, PyTorch, PySpark, Hadoop)
- Processed **30GB** of water distribution data to identify trends and patterns. Designed an LSTM forecasting model with an MAE of 1.45 and visualized results using Tableau.

- Cloud-Native API Gateway & Rate Limiter** [\[GITHUB\]](#) (Matlab, Python, Tensor Flow)
- Designed a cloud-native API gateway with authentication, request routing, rate limiting, retries, circuit breakers, and service-level observability, improving backend reliability and protecting microservices from traffic spikes.

EDUCATION

- M.S. in Computer Information Systems** | University of Massachusetts, Dartmouth | Aug 2024 - May 2026 | GPA: 4.0
- B. Tech in Electronics & Comm.** | Indian Institute of Information Technology, Sri City | Aug 2017 - May 2021 | GPA: 3.2